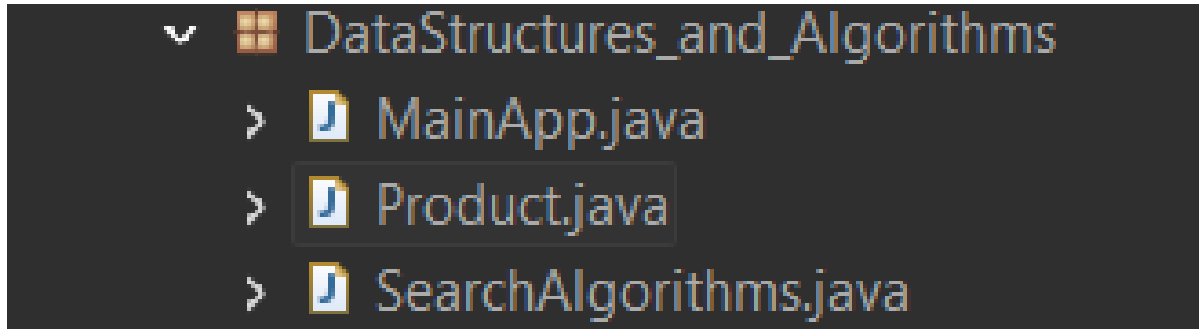


DATA STRUCTURES AND ALGORITHMS

Exercise 2 : E-commerce Platform Search Function



```
1 package DataStructures_and_Algorithms;
2
3 public class Product {
4
5     int productId;
6     String productName;
7     String category;
8
9     public Product(int productId, String productName, String category) {
10         super();
11         this.productId = productId;
12         this.productName = productName;
13         this.category = category;
14     }
15 }
16
```

```

1 package DataStructures_and_Algorithms;
2
3 public class SearchAlgorithms {
4
5     public static Product linear_search(Product[] products, String targetName) {
6         for(int i = 0; i<products.length; i++) {
7             if(products[i].productName.equalsIgnoreCase(targetName)) {
8                 return products[i];
9             }
10        }
11        return null;
12    }
13

```

```

14     public static Product binary_search(Product[] products, String targetName) {
15         int left = 0;
16         int right = products.length;
17         while(left<=right) {
18             int mid = (left+right)/2;
19             if(products[mid].productName.equalsIgnoreCase(targetName)) {
20                 return products[mid];
21             }
22             //Mid is smaller than our target --> <0
23             else if(products[mid].productName.compareToIgnoreCase(targetName)<0) {
24                 left = mid + 1;
25             }
26             //mid is greater than our target --> >0
27             else {
28                 right = mid - 1;
29             }
30         }
31         return null;
32     }

```

```

    public static void sort(Product[] products) {
        for(int i = 0; i<products.length; i++) {
            for(int j = 0; j<products.length-i-1; j++) {
                if(products[j].productName.compareToIgnoreCase(products[j+1].productName)<0) {
                    Product temp = products[j];
                    products[j] = products[j+1];
                    products[j+1] = temp;
                }
            }
        }
    }
}

```

```

1 package DataStructures_and_Algorithms;
2
3 public class MainApp {
4
5     public static void main(String[] args) {
6
7         Product[] products = new Product[5];
8         products[0] = new Product(1,"LipStick","Makeup");
9         products[1] = new Product(2,"Mobile","Electronics");
10        products[2] = new Product(3,"Doll","Toys");
11        products[3] = new Product(4,"Computer","Electronics");
12        products[4] = new Product(5,"Laptop","Electronics");
13
14        String targetName = "Laptop";
15
16        System.out.println("Linear Search");
17        Product res1 = SearchAlgorithms.Linear_search(products, targetName);
18
19        if(res1 != null) {
20            System.out.println(res1.productId + " " + res1.productName + " " + res1.category);
21        }
22        else {
23            System.out.println("Not Found");
24        }
25    }
26 }

```

```

26     SearchAlgorithms.sort(products);
27
28     System.out.println("Binary Search");
29     Product res2 = SearchAlgorithms.binary_search(products, targetName);
30
31     if(res2 != null) {
32         System.out.println(res2.productId + " " + res2.productName + " " + res2.category);
33     }
34     }else {
35         System.out.println("Not Found");
36     }
37
38
39 }
40
41 }
42

```

Problems Javadoc Declaration Console X
 <terminated> MainApp [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (21 Jun 2026, 12:39:05 pm – 12:39:05 pm) [pid: 6016]

Linear Search
 5 Laptop Electronics
 Binary Search
 5 Laptop Electronics

Exercise 7: Financial Forecasting

```

1 package DataStructures_and_Algorithms;
2
3 public class Financial_Forecasting {
4     public static double futureValue(double present, double rate, int years) {
5         if (years == 0) {
6             return present;
7         }
8         // Recursive call
9         return futureValue(present, rate, years - 1) * (1 + rate);
10    }
11    public static void main(String[] args) {
12        double result = futureValue(1000, 0.1, 3);
13        System.out.println("Future Value: " + result);
14    }
15
16 }
17

```

Problems Javadoc Declaration Console X
 <terminated> Financial_Forecasting [Java Application] C:\Program Files\Java\jdk-22\bin\javaw.exe (21 Jun 2026, 1:19:04 pm – 1:19:04 pm) [pid: 19856]

Future Value: 1331.0